Analog-1.1.44

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Problem

A signal is processed by a causal filter with transfer function $G(s)$.

For a **distortion-free output waveform**, $G(s)$ must:

(a) Provide zero phase shift for all frequencies

(b) Provide constant phase shift for all frequencies

(c) Provide linear phase shift proportional to frequency

(d) Provide phase shift inversely proportional to frequency

What is a Distortion-Free System?

A system is **distortion-free** if the output is a scaled and time-delayed replica of the input:

$$y(t) = K\,x(t - t_0), \qquad K > 0,\ t_0 \geq 0$$

Interpretation

- K — uniform scaling (amplitude change only, no shape distortion)
- t_0 — uniform time delay (all frequency components delayed equally)

Taking the Fourier transform of $y(t) = K\,x(t - t_0)$:

$$Y(j\omega) = K\,e^{-j\omega t_0}\,X(j\omega)$$

$$\therefore \quad G(j\omega) \;=\; K\,e^{-j\omega t_0}$$

Fourier Transform Approach

Step 1: Express the input as a sum of sinusoids (Fourier series/transform):

$$x(t) = \sum_n A_n \cos(\omega_n t + \phi_n)$$

Step 2: Each component passes through $G(j\omega)$:

$$|G(j\omega_n)| = K, \qquad \angle G(j\omega_n) = -\omega_n t_0$$

Step 3: Output of each component:

$$y_n(t) = K \cdot A_n \cos(\omega_n t + \phi_n - \omega_n t_0) = K \cdot A_n \cos(\omega_n (t - t_0) + \phi_n)$$

Step 4: Superposition gives:

$$y(t) = \sum_n y_n(t) = K\, x(t - t_0) \checkmark$$

Every component is delayed by the *same* t_0 — shape preserved.

Required Conditions on $G(j\omega)$

From $G(j\omega) = K\, e^{-j\omega t_0}$, two conditions follow:

Condition 1 — Magnitude (Flat):

$$\left|G(j\omega)\right| = K \quad \text{(constant for all } \omega\text{)}$$

Condition 2 — Phase (Linear):

$$\angle G(j\omega) = -\omega t_0 \quad \text{(linear in } \omega\text{)}$$

Group Delay

The group delay must be constant:

$$\tau_g(\omega) = -\frac{d}{d\omega}\angle G(j\omega) = t_0 = \text{const}$$

Constant group delay $\Leftrightarrow$ linear phase $\Leftrightarrow$ no phase distortion.

Example 1: Two-Component Signal

Let $x(t) = \cos(2\pi t) + \cos(6\pi t)$, with $K = 1$, $t_0 = 0.5\,\text{s}$.

Fourier analysis: Two components at $\omega_1 = 2\pi$ and $\omega_2 = 6\pi$.

Phase applied by $G(j\omega) = e^{-j\omega t_0}$:

$$\angle G(j\omega_1) = -2\pi \times 0.5 = -\pi \ \text{ rad}$$

$$\angle G(j\omega_2) = -6\pi \times 0.5 = -3\pi \ \text{ rad}$$

Output:

$$y(t) = \cos(2\pi(t-0.5)) + \cos(6\pi(t-0.5)) = \cos(2\pi t - \pi) + \cos(6\pi t - 3\pi)$$

$$= x(t-0.5)\checkmark$$

Observation

Both components delayed by the same $t_0 = 0.5\,\text{s}$. Output shape = input shape.

Example 2: Constant Phase (Non-Distortion-Free)

Let $x(t) = \cos(2\pi t) + \cos(6\pi t)$, but now $\angle G(j\omega) = -\dfrac{\pi}{2}$ (constant).

Time delay per component:

$$t_{\text{delay}}(\omega_1) = \frac{\pi/2}{2\pi} = 0.25 \text{ s}$$

$$t_{\text{delay}}(\omega_2) = \frac{\pi/2}{6\pi} = 0.083 \text{ s}$$

Output:

$$y(t) = \cos(2\pi t - \tfrac{\pi}{2}) + \cos(6\pi t - \tfrac{\pi}{2})$$
$$= \cos\big(2\pi(t - 0.25)\big) + \cos\big(6\pi(t - 0.083)\big)$$

Observation

Different delays for different frequencies $\Rightarrow$ shape changes $\Rightarrow$ **distortion**.

Example 3: Linear Phase Verified Numerically

$x(t) = \cos(2\pi t) + 0.4\cos(6\pi t) + 0.2\cos(10\pi t),\ K = 0.6,\ t_0 = 0.3\,\text{s}.$

Phase at each frequency:

ω (rad/s)	$\angle G = -\omega t_0$	t_{delay}	Distortion?
$2\pi \approx 6.28$	-1.885 rad	$0.3\,\text{s}$	No
$6\pi \approx 18.85$	-5.655 rad	$0.3\,\text{s}$	No
$10\pi \approx 31.42$	-9.425 rad	$0.3\,\text{s}$	No

Observation

All three components have identical delay $t_0 = 0.3\,\text{s}$.
$y(t) = 0.6\,x(t - 0.3)$ — distortion-free. ✓

Option (a): Zero Phase Shift

Claim: $\angle G(j\omega) = 0$ for all ω.

This means $G(j\omega) = K$ (purely real), implying:

$$t_0 = 0 \quad \Longrightarrow \quad y(t) = K\,x(t)$$

Problems:

- Zero delay requires the filter to be **non-causal** — the output responds instantaneously at all frequencies, which is physically unrealisable for a real causal system.
- It is an overly restrictive special case ($t_0 = 0$). Distortion-free transmission is possible with $t_0 > 0$.

Conclusion

Option (a) is **incorrect**.

Option (b): Constant Phase Shift

Claim: $\angle G(j\omega) = -\phi_0$ (a fixed constant) for all ω.

The time delay experienced by frequency component ω is:

$$t_{\text{delay}}(\omega) = \frac{\phi_0}{\omega}$$

This **varies with frequency** — low-frequency components are delayed more than high-frequency ones.

Numerical example ($\phi_0 = \pi/2$):

$$\omega = 2\pi \text{ rad/s}: \ t_0 = 0.25 \text{ s} \qquad \omega = 4\pi \text{ rad/s}: \ t_0 = 0.125 \text{ s}$$

Different delays $\Rightarrow$ waveform shape changes $\Rightarrow$ distortion.

Conclusion

Option (b) is **incorrect**.

Option (c): Linear Phase Proportional to Frequency

Claim: $\angle G(j\omega) = -\omega t_0$ (linear in ω).

Time delay for frequency ω:

$$t_{\text{delay}}(\omega) = \frac{\omega t_0}{\omega} = t_0 \quad \text{(constant!)}$$

Every frequency component is delayed by the *same* t_0 seconds, so the output waveform is identical in shape to the input:

$$y(t) = K\,x(t - t_0)\checkmark$$

Group delay: $\tau_g = t_0 = \text{const}$ ✓

Conclusion

Option (c) is **correct**. Linear phase is the necessary and sufficient phase condition for distortion-free transmission.

Option (d): Phase Inversely Proportional to Frequency

Claim: $\angle G(j\omega) = \dfrac{-K}{\omega}$.

Time delay:

$$t_{\text{delay}}(\omega) = \frac{K/\omega}{\omega} = \frac{K}{\omega^2} \quad \Longrightarrow \quad \text{frequency-dependent delay}$$

Group delay:

$$\tau_g(\omega) = -\frac{d}{d\omega}\left(\frac{-K}{\omega}\right) = \frac{K}{\omega^2} \neq \text{const}$$

Non-constant group delay $\Rightarrow$ phase distortion.

Conclusion

Option (d) is **incorrect**.

Summary

Option	Phase $\angle G(j\omega)$	Delay $t_d(\omega)$	Group Delay	Verdict
(a)	0	0 (non-causal)	0	Incorrect
(b)	$-\phi_0$ (const)	ϕ_0/ω	varies	Incorrect
(c)	$-\omega t_0$ (linear)	t_0	$t_0 = \text{const}$	**Correct**
(d)	$-K/\omega$	K/ω^2	varies	Incorrect

Answer

Option (c) — Linear phase $\angle G(j\omega) = -\omega t_0$ ensures all frequency components experience the same time delay, giving $y(t) = K\,x(t - t_0)$.

Python Code for Plot

```python
import numpy as np
import matplotlib.pyplot as plt

# Time axis
t = np.linspace(0, 4, 1000)

# Parameters: scale factor and time delay
K  = 0.6
t0 = 0.5

# Input signal (composite of two sinusoids)
x = np.sin(2*np.pi*t) + 0.4*np.sin(2*np.pi*3*t)
```

Python Code for Plot

```
# Distortion-free output: scaled and delayed
y = K*(np.sin(2*np.pi*(t-t0)) + 0.4*np.sin(2*np.
    pi*3*(t-t0)))

plt.figure(figsize=(8, 3.5))
plt.plot(t, x, 'b-',  linewidth=1.8, label=r'
    Input $x(t)$')
plt.plot(t, y, 'r--', linewidth=1.8,
          label=r'Output $y(t)=K\,x(t-t_0)$')

# Annotate the time delay
plt.annotate('', xy=(0.5+t0, 0.95), xytext=(0.5,
    0.95),
             arrowprops=dict(arrowstyle='<->',
                 color='black'))
```

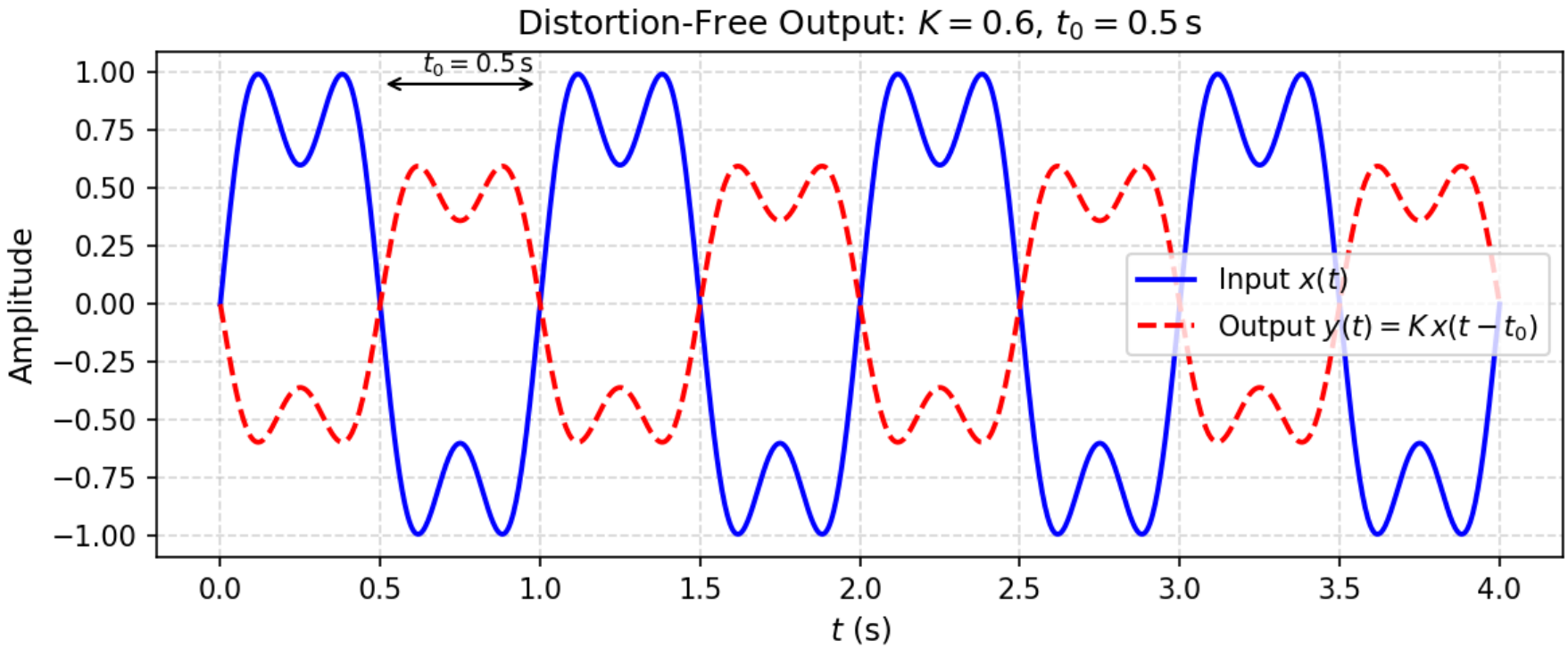
Python Code for Plot

```
plt.text(0.63, 1.0, r'$t_0=0.5\,$s', fontsize=9)
plt.xlabel(r'$t$ (s)');
plt.ylabel('Amplitude')
plt.title(r'Distortion-Free Output: $K=0.6$, $t_0
    =0.5\,$s')

plt.legend();
plt.grid(True, linestyle='--', alpha=0.5)
plt.tight_layout();

plt.savefig('plot1.png', dpi=150)
```

Plot: Distortion-Free Output



- Output has the **same shape** as the input — no distortion.
- Amplitude is scaled by $K = 0.6$ and signal is shifted right by $t_0 = 0.5$ s.